KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science
DATA SCIENCE AND VISUALIZATION (24ADI204)

INDIVIDUAL PROJECT REPORT

Youth Smoking and Drug Abuse Analysis
DATA-VIZ HACK-A-BIT 2026–2027

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Individual Work: Integrated PCA results and transformation outputs.

1. Introduction

This report presents the individual work carried out for Week 8 – Dashboard Construction in the Youth Smoking and Drug Abuse Analysis project. The work documented here focuses on the integrated PCA results and transformation outputs available in the supplied project notebook, and explains how these analytical outputs support the dashboard stage.

2. Week 8 – DASHBOARD CONSTRUCTION

2.1 Transformation Outputs

The project notebook records preprocessing outputs prepared from the cleaned Drug Consumption dataset. These include label encoding of drug-consumption categories, one-hot encoding of Gender, standardization and Min-Max scaling of the seven personality features, skewness assessment with a Yeo-Johnson transformation check, and binning of Nscore using equal-width and equal-frequency methods.

- **Label encoding:** Drug-consumption categories CL0–CL6 were mapped to numeric values 0–6. The output retained 1,885 rows and 32 columns.
- **One-hot encoding:** Gender was represented using binary columns. The output contained 1,885 rows and 33 columns.
- **Scaling:** StandardScaler and MinMaxScaler were applied to Nscore, Escore, Oscore, Ascore, Cscore, Impulsive, and SS.
- **Skewness and Yeo-Johnson:** The seven personality features had skewness values close to zero (approximately –0.0407 to 0.0728). Since none exceeded the absolute-skewness threshold of 0.5, no feature required Yeo-Johnson transformation.
- **Binning:** Nscore was grouped into Low, Medium, High, and Very High categories using both pd.cut and pd.qcut. The binned output contained 1,885 rows and 34 columns.

2.2 Integrated PCA Results

PCA was applied to the seven standardized personality features. Feature diagnostics were carried out before PCA: all seven features passed the variance-threshold check, no pair exceeded the selected correlation threshold of 0.80, and the reported VIF values ranged from 1.1508 to 1.9148. The notebook therefore retained all seven personality features for PCA.

Principal Component	Explained Variance	Cumulative Variance
PC1	31.9857%	31.9857%
PC2	25.7709%	57.7566%
PC3	12.6940%	70.4506%
PC4	9.6546%	80.1052%
PC5	8.1591%	88.2643%
PC6	6.8041%	95.0684%
PC7	4.9316%	100.0000%

The first six principal components retained 95.0684% (approximately 95.07%) of the total variance. Therefore, six components were selected to satisfy the target of retaining at least 95% of the variance. The reduced PCA output contained 1,885 rows and six columns: PC1 to PC6. The notebook also generated a PC1-versus-PC2 scatter plot and an explained-variance visualization.

2.3 Dashboard Dataset Preparation

For dashboard preparation, the notebook created a copy of the EDA-ready dataframe named dashboard_df and exported it as youth_drug_consumption_cleaned.csv. The exported dataset contained 1,885 rows and 35 columns, including demographic, personality, drug-consumption, and readable visualization-label fields such as Age_Group, Gender_Label, and Education_Level. The notebook reported zero missing values in this exported dataset.

The notebook evidences preparation and export of a dashboard-ready dataset. It does not show a completed interactive dashboard or its controls; therefore, this report records the data preparation and analytical outputs supporting dashboard construction without claiming that a finished interactive dashboard was implemented.

3. Summary of Individual Work

- Prepared and reviewed transformation outputs, including encoding, scaling, skewness assessment, and Nscore binning.
- Integrated the PCA analysis and identified six principal components retaining approximately 95.07% of variance.
- Reviewed the reduced PCA output and its supporting visualizations.
- Prepared and exported the dashboard dataset with 1,885 rows, 35 columns, and zero reported missing values.

4. Conclusion

The Week 8 work documented in this report brings together the integrated PCA results and transformation outputs and connects them to dashboard dataset preparation. PCA reduced the seven personality features to six components while retaining approximately 95.07% of the variance. The exported, labelled dataset provides a prepared input for dashboard construction. All reported results are based on the supplied project notebook.